

Milestone B, Requirements Specification

Identify the input/output elements available to the user (buttons, HDMI display, LEDs) and how the user will interact with them during the game:

Buttons - 3 buttons (left, center, right) - overlay will specify their functions:

- -1 - Play game - +1
- X - Pause - X
- New game - X - Menu

LED - Count down, Red/Green for correct incorrect amount

HDMI - Output view

Camera - Input from the user

Break down each stage of the game (Start, Round, Feedback, End, Pause) into functional requirements with timing and response constraints

Main menu - As a player, I want to see overlay with my camera view and an hint on how to start a game, as well as +1/-1 information on max player count

Start - As a player, I want to see my camera view, pause button hint and number of strikes (which should start as 0)

Round - As a player, I want to see a countdown from the LEDs with the number on the screen

Feedback - As a player, I want to see whether I was right with the use of colored LEDs (Green for correct, LED for strike), and if I was wrong number of strikes on the screen should continue

Game loop - As a player, I want the time for each switch to get smaller, the higher score I get

End - As a player, I want the game to end, when I reach strike 3

Pause - As a player, I want the game to stay in the same state when I click pause, until I click it again, where it resumed

Monitoring - As an admin, I want to see the monitoring stats (current game state, statistics, images from taking a picture?) when I access the HTTP url.

Define how each requirement can be tested (e.g., detection must occur within 1 second, etc.).

- HDMI sends proper image data with the overlays, no errors when sending + User test? No error LED shows
- After the button press, strikes should be 0-ed, in no more than 2 seconds, the overlay and view should show up
- The LEDs should start the countdown (which finishes in 2 seconds?)
- After the image, LED should show correct color in no more than a second, when this happens overlay should show different number of strikes (or finish the game)
- The time between rounds gets smaller
- State must change immediately after reaching 3rd strike, with proper information
- LED countdown state should stay consistent till unpausing, where it picks up
- HTTP is accessible and contains data with no more than 5s latency

Clarify what the system must do autonomously (e.g., timing, detection, state transitions) and what is triggered by user input

Autonomous:

- Timing for the next game state
- Saving stuff in a HTTP accessible way

- Game state transition
- Showing the webcam
- Counting the faces in an image
- Showing the overlay

User:

- Start game
- Pause game
- Unpause game

Admin:

- Accessing the HTTP with monitoring info

1. How is the selected L (max face count) input via buttons?

Left, right buttons for -1/+1

2. How is the game start triggered reliably?

Button clicked checking in another thread, which will change state?

3. How does the system manage timing transitions between rounds?

Change min/max

Time to wait = random.range(min, max)

Wait

4. How do we ensure non-repeating or unpredictable challenges?

Wait time is randomised

Number of faces is randomised

5. What visual feedback reinforces success/failure?

Green LED for success (blinking), overlay change (score up)

Red LED for failure (blinking), overlay change (into higher strike or end game state)

6. How is the game paused/resumed?

Dt is set to 0, (thus no time is subtracted/added when checking time) when paused

7. What happens if no faces are detected due to input delay or camera failure?

No output event happens, ensure heartbeat check every 10 seconds to capture errors and restart program?

8. What data is exposed by the monitoring service?

Current state (with a bit delay) - strikes, score

Past scores

Images taken? (with face recognition boxes and counts?)

Change that stuff into:

Document at least 6-8 well-structured user stories with matching testable scenarios reflecting the above problems

User Story 1 — Start Game

As a player, I want to start the game using a button so that I can begin playing immediately.

Test Scenario:

- Press the *Start button*
 - Within **≤ 2 seconds**:
 - Camera feed appears on HDMI
 - Overlay is shown
 - Strikes reset to **0**
 - System transitions to **Start state**
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User Story 2 — Adjust Max Face Count (L)

As a player, I want to adjust the maximum face count using buttons so that I can control game difficulty.

Test Scenario:

- Press **Left button** → **L decreases by 1**
 - Press **Right button** → **L increases by 1**
 - Changes reflected on HDMI **within 500 ms**
 - Value stays within allowed bounds (e.g., 1–5)
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User Story 3 — Countdown Before Capture

As a player, I want a visible countdown so that I know when the image will be captured.

Test Scenario:

- Countdown begins when round starts
 - LEDs display countdown sequence (e.g., 3 -> 2 -> 1)
 - Countdown completes within **~2 seconds**
 - Capture triggered immediately after countdown ends
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User Story 4 — Face Detection and Feedback

As a player, I want to receive feedback on whether my face count was correct so that I know if I succeeded.

Test Scenario:

- Image is captured and processed
- Within **≤ 1.5 seconds**:
 - Face count is computed
 - If correct:
 - Green LED flashes
 - Score increases
 - If incorrect:
 - Red LED flashes
 - Strike count increments

- Overlay updates accordingly
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User Story 5 — Increasing Game Difficulty

As a player, I want the game to become faster over time so that it becomes more challenging.

Test Scenario:

- After each successful round:
 - Time between rounds decreases
 - Verify:
 - Round delay reduces progressively
 - Never goes below a minimum threshold (e.g., 0.5s)
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User Story 6 — Pause and Resume

As a player, I want to pause and resume the game so that I can temporarily stop gameplay.

Test Scenario:

- Press **Pause button**
 - Game state freezes immediately
 - Countdown stops
 - Press **Pause again**
 - Game resumes from same state
 - Timing consistency maintained (no skipped time)
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User Story 7 — Game End Condition

As a player, I want the game to end after 3 strikes so that there is a clear failure condition.

Test Scenario:

- Accumulate **3 strikes**
 - System transitions to **End state immediately**
 - HDMI displays:
 - Final score
 - Game over message
 - No further rounds triggered
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User Story 8 — Monitoring System (Admin)

As an admin, I want to monitor the system remotely so that I can observe performance and game state.

Test Scenario:

- Access HTTP endpoint

- Within **≤ 5 seconds latency**, retrieve:
 - Current game state
 - Score and strikes
 - Detection stats (FPS, face count)
- System performance not degraded during monitoring